

EVALUATING COMPOSITIONAL BINDING FROM PIXELS WITHOUT ORACLE FEATURES

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ABSTRACT

Evaluations of attribute–object binding in vision–language models can be invalidated in ways that are easy to miss: the model may receive the binding through its input features, a metric may be solvable by a binding-blind content signal, or a single-modality control may look harmless only because of how the metric is constructed. We describe an evaluation setting for compositional binding in which a scene-graph generator produces minimal-change groups of two images and two captions, every group is validated by an oracle over the scene graph, and the model sees only rendered pixels and raw caption strings. Splits keep each edit orbit (all scenes with the same multisets of shapes, colours and materials, which differ only in how these are assigned to objects and positions) inside a single split, and the endpoint is restricted to binding-necessary groups, whose two scenes lie in the same orbit. We use this setting to examine a small trainable head on frozen CLIP ViT-B/32 tokens trained with a hard-negative contrastive loss. On a development panel of 64 training and 32 development groups, the head largely fits its training groups (binding-necessary training group score 0.562–0.875), but on the 16 binding-necessary development groups its group score ranges from 0.000 to 0.062 over three optimization repeats (the chance level for independent continuous scores is $1/6$), and most of its full-panel score is reproduced by a binding-blind content channel. We also show that single-modality scorers are structurally zero on 2×2 group metrics and exactly 0.5 on balanced pair AUC, so they cannot certify the absence of shortcuts. In a paired comparison on a larger panel (512 training and 128 development groups, three optimization repeats), the hard-negative loss was compared with the same loss plus a cross-group edit-consistency objective. Neither arm met a pre-fixed training-fit criterion, and training was unstable in both. The differences in favour of the edit objective appear only in runs where the baseline ended with a high final loss, so we record the comparison as unresolved and put that branch on hold. All results concern synthetic renders, a frozen encoder and development data; the held-out test splits remain unopened.

1 INTRODUCTION

Contrastive vision–language models are often described as behaving like bags of words: they retrieve images that contain the right objects and attributes but do not reliably tell apart “a red cube and a blue sphere” from “a blue cube and a red sphere” (Yuksekgonul et al., 2023; Thrush et al., 2022; Lewis et al., 2024). A common response is to train with minimal-change counterfactuals, either as hard negatives (Yuksekgonul et al., 2023; Zhang et al., 2024b;a; Awal et al., 2025) or with additional structure such as similarity-level equivariance (Wang et al., 2023) or object-centric binding modules (Assouel et al., 2025). Whether such training yields binding that transfers to attribute–object combinations or edit compositions absent from training is an empirical question, and answering it requires an evaluation that cannot be passed without binding.

This paper is about that evaluation rather than about a new method. Building it carefully exposed three failure modes that are easy to miss. First, the input can carry the answer: in our own first

version, the “image” seen by the model was one token per object assembled from the scene graph, so each token already bound its object’s shape, colour and material. A head trained on such tokens can only show how well the text side is aligned, not whether binding is learned from visual input. Second, a metric can be solvable without binding: when minimal-change groups include edits that change which colours or shapes are present, a binding-blind content channel scores well above chance (cf. Hsieh et al., 2023). Third, blind controls can look reassuring for structural reasons: on 2×2 groups a single-modality scorer ties on every comparison, so its group score is zero by construction, and on a pair metric that pairs every caption with one matching and one non-matching image its AUC is exactly 0.5. Neither number says anything about whether the benchmark contains exploitable artefacts (cf. Lin et al., 2024).

We therefore use a scene-graph generator in which every group of two images and two captions is validated by an oracle, render the scenes to RGB, and give the model only pixels and caption strings. Splits are made at the level of edit orbits, so no content multiset seen in training reappears in development or test data, and the endpoint is the subset of *binding-necessary* groups, whose two scenes have identical object and attribute multisets. Within this setting we examine the simplest reasonable learner: a small head on frozen CLIP ViT-B/32 tokens (Radford et al., 2021; Ilharco et al., 2021), trained with a hard-negative contrastive loss. We also state what a cross-group edit-consistency objective adds to that loss for the inner-product scorers used here.

Our current observations are narrow and mostly negative. On a small development panel the head fits its training groups but performs at or below chance on binding-necessary development groups, while a binding-blind content channel reproduces most of its score on the full panel. Oracle object tokens used as a control give higher, but still low, development scores, so we cannot yet separate limits of the frozen encoder from limits of the training data or the head. On a larger panel we also ran a paired comparison of the hard-negative loss with and without the edit-consistency objective under study. Neither arm fitted its training groups under the fixed update budget, and training was unstable, so that comparison is unresolved rather than negative or positive. The held-out test splits have not been opened.

Contributions. (i) A description of an oracle-validated, minimal-change evaluation setting for compositional binding with pixel-only model inputs, orbit-level splits and a metadata-defined binding-necessary endpoint (Section 3). None of these ingredients is new on its own; we document how they interact. (ii) Derivations of chance levels and tie handling for each group metric, and the structural properties of single-modality controls (Section 3, Appendices A and C). (iii) The standard bilinear identity that relates hard-negative training, the GroupMatch metric (Zhu et al., 2026) and within-group edit alignment, which isolates the cross-group part of the edit-consistency objective under study (Section 4). (iv) Development-panel observations for a hard-negative baseline on frozen CLIP tokens, with content-only, shuffle, blind and oracle-token controls. We add a pre-registered paired comparison with the edit-consistency objective, whose outcome under the pre-fixed rules is “optimization unresolved” (Section 5). (v) A metadata analysis of what orbit-level splitting does to the generated data (Appendix B).

2 RELATED WORK

Compositionality benchmarks and minimal-change training. Winoground introduced 2×2 groups with text, image and group scores (Thrush et al., 2022). ARO showed that retrieval objectives can be solved without order or binding information and proposed NegCLIP, which adds swapped captions as hard negatives (Yuksekgonul et al., 2023). CREPE separates seen and unseen compounds relative to pre-training data (Ma et al., 2023). CounterCurate and VisMin generate counterfactual image–caption pairs with generative models and fine-tune CLIP with them as hard negatives (Zhang et al., 2024a; Awal et al., 2025); CE-CLIP combines intra-modal contrast with a cross-modal ranking loss over hard negatives (Zhang et al., 2024b). Our generator plays the role of a minimal-change source, but works at the level of scene graphs, so every group is validated exactly by an oracle rather than filtered by a model or annotators. CLEVR’s CoGenT conditions hold out colour–shape palettes between training and testing (Johnson et al., 2016), and compositional zero-shot learning evaluates unseen attribute–object pairs (Nayak et al., 2023); orbit-level splitting is a stricter relative of these designs.

Metrics, blind baselines and hard positives. SugarCrepe found that text-only models beat CLIP on most earlier compositionality benchmarks and rebuilt negatives to remove such artefacts (Hsieh et al., 2023); language priors alone can solve several benchmarks (Lin et al., 2024). Test-Time Matching argues that the Winoground group score underestimates model capability and introduces GroupMatch, which asks whether the correct assignment maximizes the summed score (Zhu et al., 2026). Hard positives show that hard-negative fine-tuning can reward perturbation detection instead of understanding (Kamath et al., 2024). We use these metrics and controls, and we add the observation that in balanced 2×2 designs single-modality controls are structurally uninformative.

Binding in CLIP-style models. Lewis et al. (2024) trained compositional heads on frozen CLIP embeddings for synthetic CLEVR-like scenes and found little generalization to held-out attribute-object pairs. LABCLIP learns a linear map on frozen text embeddings with permuted negatives and argues that binding information is present uni-modally (Koishigarina et al., 2026). Gurung et al. (2025) report that neither larger batches nor explicit hard negatives make small CLIP models trained from scratch bind reliably on synthetic data. Uselis et al. (2026) relate generalization to multiplicative interactions between concepts; we read only part of that paper, and our binding channel is a multiplicative (Hadamard) interaction of this kind. OC-CLIP adds an object-centric binding module driven by parsed scene graphs (Assouel et al., 2025), building on slot-based representations (Locatello et al., 2020).

Equivariance and edit directions. EqSim constrains the similarity differences of two semantically close pairs to be consistent (Wang et al., 2023). Aligning the direction of an image-embedding change with a text-embedding change appears in StyleGAN-NADA, where frozen CLIP guides a generator (Gal et al., 2021), and PC-CLIP aligns image differences with embeddings of captions that describe the difference (Sam et al., 2025). The edit-consistency objective we study aligns image and text edit vectors across groups; we did not find it used for binding with held-out edit compositions, but our search was not exhaustive and we make no novelty claim for it.

3 PROBLEM AND EXPERIMENTAL SETTING

Scenes, edits and groups. A scene contains three objects, each with a shape (cube, sphere, cylinder, cone), a colour (red, blue, green, yellow, purple), a material (metal, rubber) and one of three horizontal slots; no two objects are identical. Four edit operations act on scenes: a *binding* edit swaps one attribute type between two objects; an *attribute* edit replaces one attribute value; an *object* edit replaces one shape; and a *relation* edit swaps the slots of two objects. Each operation carries an explicit contract on scene metadata. For example, a binding edit must leave the multisets of shapes, colours and materials unchanged while changing the multiset of (shape, attribute) pairs. Edits that violate their contract, such as swapping colours between two cubes, are rejected. A caption lists every object in one of three templates (for example “a red metal cube” or “a cube that is red and metal”) and adds one relation clause (“the X is left of the Y”). Object order, template and relation direction vary between captions and are independent of the scene layout. A *group* consists of a base scene, an edited scene and their captions under the same paraphrase choice, with member order randomized. An oracle decides whether a caption is true of a scene by searching for an injective assignment of described objects to scene objects. A group is kept only if its 2×2 truth table is the identity: each caption is true for its own scene and false for the other.

Two input conditions. The first version of this setting (v0) fed the model one token per object, computed as the sum of fixed embeddings of that object’s shape, colour, material and slot. These tokens are a function of the scene graph and already bind each object’s attributes, so any binding result obtained with them is a statement about the text side and the alignment, not about visual binding. We therefore treat them only as an *oracle object-token control*. In the pixel condition (v1), a minimal renderer draws each scene as a 224×224 RGB image (Figure 1). Shapes are drawn with distinct silhouettes and faces, colours from a fixed palette with shading, metal with high-contrast shading and a specular highlight, and rubber with matte shading. Each slot is a horizontal column, so “left of” in the scene graph is exactly the x -order in the image. Each render seed adds position and size jitter, a background level and pixel noise. Images are encoded by a frozen OpenCLIP ViT-B/32 model trained on LAION-2B (Ilharco et al., 2021; Schuhmann et al., 2022), pinned by revision and checksum. The model inputs are the final-layer class token and 49 patch tokens, each projected to

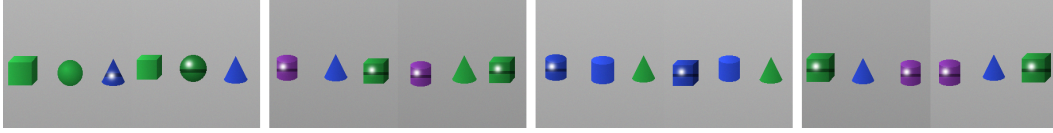


Figure 1: The first four development groups in generation order (not selected by model performance). Each pair shows the two members of one group. The pixels are those of the run output `dev_examples.png`, re-tiled into one row by `scripts/make_paper_figures.py`.

the 512-dimensional joint space. Captions are encoded as raw strings into per-token states from the start to the end token. Scene metadata is used only by the renderer, the oracle labels and the training masks; the encoder interface accepts only images and strings, and a test checks that no model scorer reads metadata.

Edit orbits and splits. Binding and relation edits generate a group action on scenes: transpositions of colours, of materials or of slots between objects generate all permutations of each attribute type, with shapes attached to object identities. Ignoring the validity constraint on intermediate scenes, the *edit orbit* of a scene under this action is the set of valid scenes with the same multisets of shapes, colours and materials (positions always fill the three slots), and the split key is exactly this triple of multisets. Attribute and object edits change these multisets and so move a scene to a different orbit. Data v0 prevented leakage at the level of scenes and caption contents, so all render variants and paraphrases of a scene or caption stayed in one split. Data v1 adds the orbit as a leakage unit: every orbit is assigned to one split by a hash of its key, base scenes are drawn only from the split’s orbits, and an edit is kept only if its result stays in an orbit owned by the same split. This restriction changes the distribution of accepted content edits relative to unrestricted sampling; Appendix B reports the measured proposal and acceptance counts instead of assuming uniformity. Data v1 uses a fresh seed and the v0 split sizes (2,000 training, 300 development, 300 calibration and 400, 400, 400 and 400 groups in the four test splits). The v0 test splits were scored during an early technical smoke run and are treated as development material; no model has been scored on any v1 test split.

Binding-necessary endpoint. A group is *binding-necessary* (BN) when all of its edits are binding or relation edits. Equivalently, both of its scenes lie in the same orbit, so the two images contain the same objects and attribute values and differ only in their assignment or arrangement. The subset is defined by metadata alone and never by model scores. It is the primary endpoint because on groups with content edits a binding-blind signal can succeed: in the v0 smoke run (oracle tokens, first 100 groups of the v0 composition split), the content-only channel of each trained head reached GroupMatch 0.650–0.760, against a chance level of 0.5.

Head and scorer. The trainable head is shared by all conditions. For a set of image tokens $\{v_k\}_{k=1}^n$ it outputs a content part $c = W_c \frac{1}{n} \sum_k v_k$ and a binding part $b = \frac{1}{n} \sum_k (Av_k) \odot (A'v_k)$. For text tokens $\{u_t\}_{t=1}^L$ the binding part is $\frac{1}{L} \sum_t \sum_{\delta \in \pm\{1..5\}} \alpha_\delta (Bu_t) \odot (B'u_{t+\delta})$ with learned offset weights α_δ . The score is the inner product $s(I, T) = \langle [c_I; b_I], [c_T; b_T] \rangle$. With 8 content and 8 binding dimensions and 512-dimensional inputs, the head has 24,586 parameters; the frozen encoder has 151,277,313. Tokens are standardized per dimension with statistics of the training panel. With the additive oracle tokens of v0, the content part is exactly invariant to binding and relation edits by construction. With CLIP tokens this invariance no longer holds, and we measure it instead of assuming it (Section 5).

Metrics, ties and chance levels. For a group with score matrix $S_{ij} = s(I_i, T_j)$, the Winoground *text* score requires $S_{00} > S_{01}$ and $S_{11} > S_{10}$, the *image* score requires $S_{00} > S_{10}$ and $S_{11} > S_{01}$, and the *group* score requires both (Thrush et al., 2022). *GroupMatch* requires $S_{00} + S_{11} > S_{01} + S_{10}$ (Zhu et al., 2026). With a meaning-preserving paraphrase T'_i of each caption, *aug* counts images for which both T_i and T'_i beat T_{1-i} , and *aug-group* requires this for both images (after Kamath et al., 2024). The *pair AUC* treats the four image–caption pairs of each group as a binary ranking problem labelled by the oracle. All comparisons are strict with a relative tolerance of 10^{-9} , so ties, including round-off differences between mathematically equal scores, count as failures. AUC assigns tied scores their average rank. For independent continuous scores the chance levels are $1/4$

(text, image), 1/6 (group), 1/2 (GroupMatch), 1/3 (aug), 1/9 (aug_group) and 0.5 (pair AUC). Each is derived separately in Appendix A; we do not transfer the $1/k!$ formula for GroupMatch to other metrics. A Monte-Carlo check on the development panel gives 0.168 for group and 0.500 for GroupMatch.

Controls. We report an oracle scorer (metadata truth, an upper bound for the evaluation code), zero-shot CLIP cosine scores, a text-only n-gram logistic regression and an image-only logistic regression on pooled CLIP features, the content-only and binding-only channels of the trained head, the trained head with development images or captions shuffled across groups, and the same head trained on oracle object tokens. On 2×2 groups a single-modality scorer produces two identical rows or columns. Every comparison therefore ties, and its text, image, group and GroupMatch scores are zero by construction. Its pair AUC is exactly 0.5 because every caption, and every image, appears once with each label (Appendix C). These values are properties of the design, not evidence that the benchmark is free of artefacts. For artefacts, the informative checks are the content-only channel, the shuffles, and classifiers that try to tell base from edited scenes.

4 OBJECTIVES UNDER STUDY

Hard-negative contrastive loss. A batch contains B groups, hence $2B$ images and $2B$ captions. With S the $2B \times 2B$ score matrix, the loss $\mathcal{L}_{\text{hn}}$ is the average of row-wise (image $\rightarrow$ caption) and column-wise (caption $\rightarrow$ image) cross-entropies with diagonal targets. Every off-diagonal pair that the oracle marks as true is removed from the denominators, so it is never used as a negative. The group partner of each item is kept as a negative (the hard negative). This is the NegCLIP-style use of minimal-change partners (Yuksekgonul et al., 2023; Awal et al., 2025), applied here to a small head on frozen features.

Remark 1 (a standard bilinear identity). *Assume the score is an inner product $s(I, T) = \langle \phi(I), \psi(T) \rangle$, where ϕ depends only on the image and ψ only on the caption. Per-input normalization, as in cosine scores, is allowed; normalization or attention that depends on the pair is not. Then, for a 2×2 group,*

$$\begin{aligned} S_{00} + S_{11} - S_{01} - S_{10} &= \langle \phi(I_0), \psi(T_0) \rangle + \langle \phi(I_1), \psi(T_1) \rangle - \langle \phi(I_0), \psi(T_1) \rangle - \langle \phi(I_1), \psi(T_0) \rangle \\ &= \langle \phi(I_0) - \phi(I_1), \psi(T_0) - \psi(T_1) \rangle, \end{aligned} \tag{1}$$

by bilinearity alone. Both the head of Section 3 and zero-shot cosine CLIP satisfy the assumption.

Equation equation 1 is elementary; we state it because it fixes what several signals share. GroupMatch is exactly the event that the right-hand side is positive. The two within-group row margins of the hard-negative loss, $S_{00} - S_{01}$ and $S_{11} - S_{10}$, sum to the same quantity. Aligning the image edit vector $\phi(I_1) - \phi(I_0)$ with the caption edit vector $\psi(T_1) - \psi(T_0)$ of the same group rewards the same inner product, up to normalization. None of this holds automatically for scorers with pair-dependent computation, such as cross-attention heads, and we do not apply it to them.

The edit-consistency objective under study. For each group g in a batch let $\Delta_g^I = e(I_1^g) - e(I_0^g)$ and $\Delta_g^T = e(T_1^g) - e(T_0^g)$, using the same member order in both modalities, where e is the concatenated head output. The objective $\mathcal{L}_{\text{edit}}$ is a symmetric cross-entropy over the $B \times B$ matrix $C_{gh} = \cos(\Delta_g^I, \Delta_h^T) / \tau$ with diagonal targets and $\tau = 0.1$. Norms include a small ϵ , so zero edit vectors give uniform logits and a loss of $\log B$: unlike a squared error between edit vectors, zero is not a minimizer. The diagonal entries reward the normalized within-group quantity of Remark 1 and therefore overlap with the hard-negative loss. The off-diagonal entries are the additional signal: an image edit must be closer in direction to its own caption edit than to the caption edits of other groups in the batch. The objective assumes that different groups in a batch carry different edits; edit-level false negatives are not masked. The combined loss is $\mathcal{L}_{\text{hn}} + \lambda \mathcal{L}_{\text{edit}}$.

Scale. On the first training batch of the small panel, before training and with $\lambda = 1$, the ratio $\|\nabla \mathcal{L}_{\text{edit}}\| / \|\nabla \mathcal{L}_{\text{hn}}\|$ was 11.7–14.0 across seeds; after hard-negative training it was 0.27–0.55. We report these ratios only to show that λ must be small for the two terms to be on comparable scales. A large gradient says nothing about usefulness.

Table 1: Small development panel (64 training / 32 development groups, 16 binding-necessary). Entries for trained heads list seeds 0/1/2, which are optimization repeats on the same panel. BN: binding-necessary subset; “all”: all four edit kinds. Chance for independent continuous scores: group 1/6, GroupMatch 1/2, pair AUC 0.5. Single-modality rows are zero or 0.5 by construction (Section 3). Source: `runs/pixel_baseline_v1_20260926T161957Z/metrics.json`.

Scorer	BN group	BN GroupMatch	BN pair AUC	all: group
Oracle (metadata)	1.00	1.00	1.00	1.00
Zero-shot CLIP	.000	.688	.497	.312
Text-only (n-gram LR)	.000	.000	.500	.000
Image-only (CLIP LR)	.000	.000	.500	.000
Pixel head (hard-neg.)	.062/.000/.000	.625/.500/.438	.508/.500/.508	.375/.156/.188
content channel	.000/.000/.000	.562/.562/.438	.506/.504/.494	.219/.156/.188
binding channel	.125/.062/.000	.625/.625/.562	.515/.537/.511	.062/.062/.000
images shuffled	.000/.000/.000	.312/.312/.375	.488/.496/.488	.000/.062/.031
captions shuffled	.000/.000/.000	.375/.500/.750	.502/.487/.494	.062/.000/.000
train-panel fit	.750/.562/.875	1.00/1.00/1.00	.716/.621/.657	.844/.781/.938
Oracle object tokens	.312/.125/.188	.812/.750/.750	.619/.548/.569	.312/.094/.188

5 EXPERIMENTS AND OBSERVATIONS

All numbers are produced by one exporter from the raw run files and inserted as macros (Appendix F). Every experiment uses data v1, frozen features, one CPU thread and development data only. Seeds are optimization repeats on the same panel, not independent samples.

5.1 RENDERER AND DATA AUDITS

Before any model was run, an audit-only pixel decoder recovered slot, shape, colour and material of every object from the rendered images. It flagged a renderer error: the specular highlight of metal cones fell outside the cone’s narrow top, so those cones looked like rubber. After the fix, the decoder found 0 mismatches on all 288 images of the small panel. The decoder is a hand-written check written with the renderer, so it shows self-consistency, not independent visual verification. Data v1 passed all structural and leakage audits (0 failures). Logistic-regression detectors that try to tell the base from the edited member reached AUC 0.508 (caption n-grams) and 0.511 (oracle tokens) on 1,200 development and calibration items. A detector on pooled CLIP image features reached 0.47 with a 95% bootstrap interval of [0.31, 0.61] on the 64 panel images. That interval is wide, and none of these linear detectors rules out nonlinear traces.

5.2 SMALL DEVELOPMENT PANEL: HARD-NEGATIVE BASELINE

The baseline was trained with a single setting fixed before the run: learning rate 0.01, weight decay 10^{-4} , 300 steps of 16 groups, logit scale 1, and seeds 0, 1 and 2. It used the first 64 training and first 32 development groups of data v1. Half of the development groups (16) are binding-necessary. Table 1 reports the results.

Observations. (1) The evaluation path behaves as intended. The oracle scores 1.000, and shuffling development images or captions across groups drives the BN group score to 0.000 and 0.000 respectively in every seed. (2) The head largely fits its training groups: on the 16-group fit subset its group score was 1.000, and on the full training panel its BN group score was 0.562–0.875. The fit is surface-bound, however: on the fit subset, the aug-group score with meaning-preserving paraphrases was only 0.375. (3) On binding-necessary development groups the head is at or below chance: its group score is 0.000–0.062, its GroupMatch 0.438–0.625 and its pair AUC 0.500–0.508. (4) Most of the head’s full-panel group score (0.156–0.375) is reproduced by its content channel alone (0.156–0.219). Zero-shot CLIP shows the same pattern: 0.312 on all kinds but 0.000 on the BN subset, although its BN GroupMatch is 0.688. (5) The oracle object-token control, whose input already binds attributes, reaches BN development group scores of only 0.125–0.312 after fitting its training groups perfectly (1.000). With 64 training groups, therefore, the head and the data size

Table 2: Paired comparison on 512 training and 128 development groups (64 binding-necessary). Seeds 0/1/2 are paired optimization repeats on the same panel. Chance for independent scores: group 1/6, GroupMatch 1/2, pair AUC 0.5. Source: runs/paired_v1_20260926T230018Z/paired_summary.json.

Seeds 0/1/2	A: hard-neg.	B: + edit-cons.	B−A
Dev BN group score	.094/.016/.000	.000/.125/.234	−.094/.109/.234
Dev BN GroupMatch	.688/.641/.656	.609/.812/.719	−.078/.172/.062
Dev BN pair AUC	.520/.503/.501	.506/.549/.574	−.014/.045/.073
Dev binding-only group	.156/.000/.000	.000/.156/.469	−.156/.156/.469
Dev relation-only group	.031/.031/.000	.000/.094/.000	−.031/.062/.000
Dev all-kinds group	.367/.289/.250	.297/.383/.461	−.070/.094/.211
Dev all-kinds GroupMatch	.844/.820/.828	.805/.906/.859	−.039/.086/.031
Train BN group (fit)	.535/.152/.066	.094/.746/.703	−.441/.594/.637
Final task loss	0.51/1.83/7.18	0.91/0.25/0.29	0.40/−1.58/−6.89

already limit generalization, before any limit of the encoder comes into play. We cannot separate these factors on this panel, and with 16 BN groups we make no significance claim in either direction.

5.3 ENCODER SENSITIVITY DIAGNOSTICS

For each development group we compared the cosine distance between the pooled CLIP embeddings of the two members with the distance between two renders of the same member under different seeds. The median ratio was 4.04 for object edits, 2.27 for attribute edits, 1.49 for binding edits and 1.04 for relation edits. Relation swaps therefore move the pooled embedding about as much as rendering nuisance does. The content channel is no longer invariant: for binding-necessary pairs rendered with a shared seed, the median relative content change $\|c(I_1) - c(I_0)\|/\|c(I_0)\|$ was 0.23–0.28 across seeds, whereas with v0 oracle tokens it was zero by construction. The invariance of v0 was a property of its additive inputs, not something the head learned.

5.4 PAIRED COMPARISON WITH THE EDIT-CONSISTENCY OBJECTIVE

Design, fixed before the run. The panel contains the first 512 training groups (427 orbits) and 128 development groups (74 orbits). The development groups start after the 32 groups already scored in the small-panel stage. The two panels share 0 orbits, and 64 of the development groups are binding-necessary. Arm A uses the hard-negative loss. Arm B adds $\lambda \mathcal{L}_{\text{edit}}$ with $\lambda = 0.01$, the smallest non-zero value of the earlier pre-registered grid. We disclose that this grid was set after observing gradient scales, and λ was not changed after any development result was seen. Both arms share initialization, batch order, supervision, false-negative masks, cached encoder features, optimizer (AdamW, learning rate 0.01, weight decay 10^{-4}) and 1,500 updates of 16 groups. Seeds 0, 1 and 2 are paired optimization repeats, and the development panel is scored only after the final update. At initialization the edit gradient was 13.1 times the task gradient (mean over the three first batches), so λ scales it to about 0.13 of the task gradient. Encoding the 1,408 panel images and their captions took 259.3 CPU seconds, and the audit decoder found 0 mismatches.

Outcome under the pre-fixed rules. Neither arm met the fit criterion, a seed-mean binding-necessary training group score of at least 0.75: A reached 0.251 and B 0.514 (Table 2). The pre-registered verdict is therefore OPTIMIZATION_OR_INPUT_UNRESOLVED. Training was unstable in both arms. The task loss oscillated during training, and the largest logged gradient norm per run ranged from 77.3 to 427.7. Arm A finished seed 2 at a task loss of 7.18, above its initial value, and seed 1 at 1.83. Arm B finished at 0.91, 0.25 and 0.29.

What the difference does and does not show. On binding-necessary development groups, the edit-consistency arm scored higher in 2 of 3 seeds. The mean paired difference in group score is 0.083. Its group-level bootstrap interval is [0.036, 0.135], but that interval treats the three runs as fixed and does not reflect their optimization variance. The per-seed differences in Table 2 are positive in the seeds where the baseline ended with a high final loss, and negative in the seed where it did not. We therefore do not read them as added utility of the objective. Both arms also sit below

the independent-score chance level on the group score (A: 0.036, [0.010, 0.062]; B: 0.120, [0.078, 0.167]) and above it on GroupMatch (A: 0.661; B: 0.714). A scorer whose preferences are correlated across the two captions, for instance one that prefers the same image for both, can fall below 1/6 on the group score while still matching better than 1/2, so the two metrics must be read together. Zero-shot CLIP on the same subset reaches 0.031 (group) and 0.594 (GroupMatch). Following the pre-fixed rules, the edit-objective branch is put on hold. No stabilization, encoder change, data increase or new loss was started on the basis of this result.

6 DISCUSSION AND LIMITATIONS

What the observations support. The setting does what it was built to do: model inputs are produced from pixels and caption strings alone, the evaluation path passes its oracle and shuffle checks, and the binding-necessary subset removes the part of the metric that a binding-blind channel can solve. On this subset, a hard-negative head on frozen CLIP tokens fits its training groups on a small panel but does not transfer to held-out orbits. On a larger panel, under a single fixed optimizer setting, neither the baseline nor the edit-consistency variant fitted its training groups, and training was unstable. This is a statement about one encoder, one head, one training setting and small panels. It is neither a claim that the encoder lacks binding information nor a claim that hard negatives fail in general.

Limitations. *Optimization.* The paired comparison used one fixed learning rate, no learning-rate schedule, no gradient clipping and unnormalized inner-product scores, and it was evaluated only at the final update. Its per-seed differences are confounded with loss spikes, and its group-level bootstrap interval ignores run-to-run variance. *Synthetic renders.* The images are stylized renders of four shapes, five colours and two materials; nothing here concerns natural images, and we did not test whether CLIP maps the words “metal” and “rubber” onto our rendering. *Frozen encoder.* Only the head is trained, so no result speaks to what fine-tuning the encoder would do. Only final-layer tokens of one checkpoint were used, and other layers or token choices were not examined. *Small panels and dependent repeats.* The development panels are small, and the three seeds share the same data, so their spread reflects optimization variance only. Non-significant differences are not evidence of equivalence. *Self-checked renderer.* Our decoder shows that the renderer is internally consistent, not that an independent observer would read the scenes the same way. *Data-generation bias.* Keeping orbits within splits requires restricting content edits to owned orbits, which changes their distribution. Small splits also differ from training in attribute composition (Appendix B). *Unopened test.* No v1 test split has been scored, so no result here is confirmatory. *Prior art.* Each ingredient of the setting and both objectives have close precedents (Section 2); we have not established novelty for any of them. *Authorship.* The code and this text were produced by an AI system and have not yet been reviewed by a human author.

7 CONCLUSION

A binding evaluation is only as informative as the weakest of its inputs, metrics and controls. By removing oracle features from the model input, splitting at the level of edit orbits and scoring a metadata-defined binding-necessary subset, we obtained a setting in which a simple hard-negative head on frozen CLIP tokens can be seen to fit its training groups without transferring to held-out orbits, and in which blind controls are recognized as structurally uninformative. The paired comparison with the edit-consistency objective under study did not resolve whether it adds anything to the hard-negative loss. Neither arm fitted its training groups under the fixed update budget, both training runs were unstable, and the per-seed differences track which baseline runs ended with a high final loss. A useful next step would stabilize optimization under a new pre-registered configuration before comparing objectives again. The v1 test splits remain unopened.

AI USE STATEMENT

This working draft, the code base and every experiment it reports were produced by an AI coding assistant (Anthropic’s Claude, used through Claude Code) acting on a researcher’s written instructions. The assistant designed and implemented the generator, renderer, head, losses and evaluation

code; ran all experiments; verified bibliographic metadata through arXiv’s BibTeX export and paper pages; and wrote this text. AI sub-agents searched and read related work, and their reports were partially re-checked by the main assistant. **HUMAN_REVIEW_PENDING:** no human author has yet checked the code, the numbers or the text, and no human is named as having verified anything. The draft must not be submitted before that review. The repository file `AI_USAGE.md` lists the AI’s design decisions and the errors it found and corrected.

REPRODUCIBILITY STATEMENT

Section 3 and Appendices B–E describe the data, splits, inputs, head, objectives and metrics. Every number in the text is generated from raw run files by `scripts/export_paper_numbers.py`, which writes a provenance table mapping each macro to a file and field (Appendix F). Run directories contain configuration, logs and manifests with code, configuration and data hashes, the environment, wall and CPU time, and peak memory. The encoder checkpoint is pinned by revision and SHA-256, and Python packages are pinned in `requirements-pixel.lock`.

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A CHANCE LEVELS OF THE GROUP METRICS

Let the four scores of a group, $S_{00}, S_{01}, S_{10}, S_{11}$, and the paraphrase scores S'_0, S'_1 be independent draws from one continuous distribution, so ties have probability zero. These are properties of random scores, not of any trained model. Each metric is derived separately.

Text and image scores. The text score needs $S_{00} > S_{01}$ and $S_{11} > S_{10}$. These two events involve disjoint pairs of independent exchangeable variables, so the probability is $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. The image score is analogous.

Group score. All four inequalities hold iff $\min(S_{00}, S_{11}) > \max(S_{01}, S_{10})$, that is, iff the two diagonal entries are the two largest of four exchangeable values. All $\binom{4}{2} = 6$ pairs are equally likely to be the top two, so the probability is $1/6$.

GroupMatch. $X = S_{00} + S_{11}$ and $Y = S_{01} + S_{10}$ are independent and identically distributed, so $P(X > Y) = 1/2$. For $k \times k$ groups Zhu et al. (2026) give $1/k!$ for GroupMatch and $(k-1)/(2k-1)!$ for the group score; we use only $k = 2$ and do not carry either formula over to the other metrics.

aug. For image i , the other caption must score below both S_{ii} and S'_i , that is, be the smallest of three exchangeable values: probability $1/3$.

aug-group. The two images use the disjoint triples (S_{00}, S'_0, S_{01}) and (S_{11}, S'_1, S_{10}) , so the two events are independent and the probability is $1/9$.

Pair AUC. Under random scores, the expected AUC is $1/2$.

B SPLITS AND METADATA STATISTICS

Orbit ownership is decided by a hash of the orbit key and the data seed, with fractions 0.50 (training), 0.075 (development), 0.075 (calibration), 0.10 for each of the i.i.d., held-out-pair and composition test splits, and 0.05 for the held-out-template test split (`configs/data_v1.json`). For each split, the generator draws a base scene until its orbit is owned. It then samples an edit of the scheduled kind among the applicable edits whose result stays in an owned orbit, and it keeps the group only if the oracle truth table, the operation contract, the held-out-pair rule and the scene/caption-content registry all pass. Edit kinds follow a fixed cycle, so accepted kinds are balanced within each split. Content edits (attribute and object edits) are the only edits that can leave an orbit, and only they are affected by the ownership filter. Table 3 reports the metadata of every data-v1 split, recomputed by `scripts/analyze_split_semantics.py` from the generator’s own bookkeeping. No model is involved and no test score is opened. Of the 2,440 orbits of three-object scenes, each split owns a hashed subset. “Content filter keep” is the fraction of applicable attribute and object edits whose result stays in an orbit owned by the same split. Binding and relation edits never leave their orbit, and all of them were kept.

Table 3: Data-v1 split metadata (source: `runs/split_semantics_v1_20260926T230848Z/split_semantics.json`). “Attempts” counts every proposal, including rejected base scenes; “accept” is groups divided by attempts; “orbits” is orbits owned, “used” orbits occurring in the split; “keep” is the content-edit filter keep rate; BN is the binding-necessary share.

Split	groups	attempts	accept	orbits	used	keep	BN
train	2000	6743	29.7%	1241	758	50.1%	50.0%
dev	300	7063	4.2%	173	97	10.4%	50.0%
calib	300	19334	1.6%	176	91	4.7%	50.0%
test i.i.d.	400	6858	5.8%	241	121	10.6%	50.0%
test held-out pairs	400	14309	2.8%	243	146	10.0%	50.0%
test composition	400	25297	1.6%	245	149	10.9%	30.0%
test held-out template	400	80837	0.5%	121	68	3.4%	50.0%

Consequences. (i) Accepted content edits are not a uniform sample of applicable edits. In the development split, 168 of 1,484 applicable colour replacements and 152 of 1,684 applicable shape replacements survived the ownership filter, while 305 of 305 colour swaps did. Calibration and held-out-template splits keep even fewer content edits (4.7% and 3.4%). (ii) Small splits own few orbits, so their attribute composition differs from training. The total-variation distance to the training colour distribution is 0.12 for the development split. In the paired development panel green and purple have frequencies 0.26 and 0.29, against red and blue at 0.12 and 0.12; in the training panel the figures are 0.17, 0.26, 0.19 and 0.19. (iii) Edit kinds are balanced by construction: half of each single-edit split is binding-necessary, and 50.0% of the paired development panel is. (iv) The analysis was first run on an uncommitted tree and then re-run on the committed code. The two outputs agree except

for one total-variation value that differs in the last floating-point digit because of summation order. Both runs are kept.

C STRUCTURAL PROPERTIES OF SINGLE-MODALITY SCORERS

Let a text-only scorer assign $S_{ij} = f(T_j)$. Then $S_{00} = S_{10}$ and $S_{01} = S_{11}$.

- The text score needs $f(T_0) > f(T_1)$ and $f(T_1) > f(T_0)$, which is impossible.
- The image score needs $S_{00} > S_{10}$, a tie.
- GroupMatch compares $f(T_0) + f(T_1)$ with itself.
- aug-group needs the text-score inequalities again.

With strict comparisons all four are therefore zero for every f . The per-image aug score can still reach $1/2$, so it is not blind-proof. For the pair AUC, the four pairs of a group carry labels given by the identity truth table. Each caption T_j appears once with label 1 (with I_j) and once with label 0 (with I_{1-j}), with the same score $f(T_j)$. The multisets of positive and negative scores are therefore identical. The Mann–Whitney statistic, with ties counted as $1/2$, then equals $1/2$ exactly. The image-only case is symmetric. These identities hold for any single-modality scorer, including ones that exploit real artefacts, so the corresponding table entries carry no information about artefacts.

D SMALL-PANEL RESULTS OVER ALL EDIT KINDS

Table 4: Small development panel, metrics over all four edit kinds (same run, rows and conventions as Table 1). These include groups that a binding-blind content signal can solve.

Scorer	all: group	all: GroupMatch	all: pair AUC
Oracle (metadata)	1.00	1.00	1.00
Zero-shot CLIP	.312	.844	.583
Text-only (n-gram LR)	.000	.000	.500
Image-only (CLIP LR)	.000	.000	.500
Pixel head (hard-neg.)	.375/.156/.188	.812/.750/.719	.579/.571/.572
content channel	.219/.156/.188	.781/.781/.719	.579/.570/.569
binding channel	.062/.062/.000	.500/.562/.562	.507/.514/.513
images shuffled	.000/.062/.031	.312/.406/.438	.498/.493/.500
captions shuffled	.062/.000/.000	.469/.562/.656	.519/.490/.497
train-panel fit	.844/.781/.938	1.00/1.00/1.00	.758/.725/.745
Oracle object tokens	.312/.094/.188	.906/.781/.875	.620/.571/.576

E PLANNED EXPERIMENTS (NOT RUN)

None of the following has been executed; no result in the main text depends on them.

1. **Stabilized paired comparison.** The paired comparison of Section 5.4 was not repeated with a lower learning rate, a schedule, gradient clipping, normalized scores or checkpoint averaging. Any such change would need a new configuration fixed before running.
2. **Test evaluation.** No v1 test split has been rendered, encoded or scored.
3. **Natural images and encoder fine-tuning.** Not attempted. No natural-image data was used, and the encoder stays frozen.
4. **Other encoders, layers or token choices.** Not attempted (single encoder, final layer).
5. **Calibrated pair accuracy.** Requires a calibration panel, which has not been fixed.
6. **Edit-loss hyperparameter sweep.** The earlier 12-setting sweep was withdrawn and not run.

F PROVENANCE AND CORRECTIONS

`scripts/export_paper_numbers.py` reads the raw run files, writes every number as a macro to `paper/generated/numbers.tex`, builds the tables from the same values, and records file, field and derivation rule per macro in `paper/generated/provenance.tsv`. Corrections made during the project are recorded rather than hidden:

- *Renderer*. The metal-cone highlight error was found by the audit decoder and fixed before any model was run.
- *First pixel-stage run*. It was executed on an uncommitted working tree. A re-run at the committed code gave identical metrics, decisions and parameter hashes, and both runs are kept.
- *Manifest*. The “dirty” flag in the reference run’s manifest is a false positive caused by a parsing bug, since fixed; the recorded source-tree hash identifies the code.
- *Installation*. A first package-installation attempt failed because a timing utility was missing; its log is kept.